FairDermNet: Diagnosing and Correcting Silent Fairness and Quantization Failures in an On-Device Skin Lesion Classifier

Sanjana T. G.
Department of Computer Science and Technology
University of Cambridge
sanjanatg2126@gmail.com

Abstract—DermaLens is an 8-class, fairness-aware, on-device skin lesion risk classifier (MobileNetV3-Large backbone, GeM pooling, Fitzpatrick-tone auxiliary head) exported to TFLite for offline mobile inference. During a code audit prior to deployment we discovered three independent, silently-compounding defects: (1) a stale binary risk-label mapping that mislabeled the majority benign class as malignant in evaluation code, (2) a broken Individual Typology Angle (ITA°) skin-tone estimation call that fabricated tone labels for 97% of training data, causing the fairness-aware loss to train on a degenerate single-tone distribution, and (3) catastrophic accuracy collapse (73.7%→3.3% top-1) under full INT8 post-training quantization, traced to an unquantizable learned power operation in Generalized Mean (GeM) pooling. We present root-cause diagnosis, targeted fixes, and quantitative before/after evidence for each defect, and report a corrected deployment pipeline (dynamic-range quantization, 5.51 MB) validated end-to-end on-device. Score-level analysis of the out-of-domain failure shows the malignancy score carries no measurable ranking signal for FST V–VI images (AUC 0.484 [0.382, 0.585]), traced to a spurious, capture-associated actinic-keratosis response that is invariant to skin tone within DDI and cannot be corrected post hoc – a representational domain gap rather than a threshold-calibration failure. We show that naïve re-training does not address fairness defects rooted in label corruption, and argue for data-lineage auditing as a mandatory pre-deployment step for fairness-aware medical ML.

Index Terms—skin cancer classification, algorithmic fairness, model quantization, TinyML, on-device inference, dermoscopy

I. Introduction

Skin cancer screening models deployed on consumer smartphones must satisfy three simultaneous constraints: diagnostic accuracy, equitable performance across Fitzpatrick skin tones I–VI, and a small enough footprint (target < 6 MB) to run fully offline. DermaLens targets an 8-class taxonomy (Melanoma, BCC, SCC, AK, Nevi, BKL, DF, VASC; malignant = classes 0–3) trained on HAM10000 [5], ISIC 2019 [6], PAD-UFES-20 [7], and evaluated on the expert-labelled Diverse Dermatology Images (DDI) dataset for fairness. This paper documents a structured audit that found the system's fairness evaluation and quantized export were both silently broken, despite training and top-line test accuracy appearing normal.

II. Related Work

Skin-lesion classifiers are built on HAM10000 [5], ISIC 2019 [6] and PAD-UFES-20 [7], and have repeatedly been shown to degrade on darker skin: Daneshjou et al. evaluate publicly released dermoscopy models on the expert-labelled DDI test set [8] and report the sensitivity disparities that motivate our evaluation. Fairness-aware training typically penalizes group-conditioned error imbalance [2]; attribution tools such as Grad-CAM [1] are the standard instrument for inspecting what such models attend to; and GeM pooling [9] is a common retrieval-era pooling whose quantization behavior has, to our knowledge, not previously been analyzed. Obermeyer et al. show that label choices can encode bias even when aggregate accuracy looks acceptable [3], and Finlayson et al. catalogue the dataset-shift hazards behind silently degrading clinical models [4] – the mechanisms behind Defects 1 and 2 here. Unlike prior work that primarily reports disparities, this paper traces three specific silent defects end-to-end and shows that a four-layer chain of post-hoc corrections cannot repair the resulting out-of-domain failure.

III. System Overview

FairDermNet combines a MobileNetV3-Large backbone with a Fairness Attention Module and Generalized Mean (GeM) pooling [9], and a dual-head output: an 8-class risk head and a Fitzpatrick-tone auxiliary head, trained jointly with loss

$$\mathcal{L} = \alpha\,\mathcal{L}_{cls} + \beta\,\mathcal{L}_{fitz} + \gamma\,\mathcal{L}_{eo}$$

where $\mathcal{L}_{eo}$ penalizes the equalized-odds gap [2] across tone groups. A provenance audit (audit_step0_1.py, September 2026) verified that the training-time equalized-odds term (FairnessAwareLoss, N_MALIGNANT = 4) and the per-epoch fairness signal both operated on the full malignant set {MEL, BCC, SCC, AK}; the deployed checkpoint therefore reflects optimization against the complete four-class malignancy definition, and no retraining was required. The model is exported PyTorch → ONNX → TFLite for on-device Flutter inference.

IV. Defect 1: Stale Binary Risk-Label Mapping

Evaluation and threshold-calibration code derived a binary malignant/benign label via a hardcoded set membership test left over from a prior 9-class taxonomy (class_idx in {0,1,2,3,4}) instead of the canonical class_idx_to_binary() taxonomy function. Because Melanocytic Nevi (index 4) is the single largest class (53% of training rows) and is unambiguously benign, this silently mislabeled it malignant in every consumer of the derived label. Training loss was unaffected (it consumes class labels directly), but three downstream artifacts were corrupted:

- DDI fairness evaluation used a derived label that, because DDI rows carry a placeholder class_idx=-1, evaluated to a constant 0 for all 656 samples, causing an AUC computation failure and previously reported equalized-odds gap figures to originate from a stale code path, not the current model.
- Clinical threshold calibration (Cell 9c) would have included Nevi probability mass in the malignancy score.
- GradCAM ground-truth display referenced a non-existent dataframe column.

Fix. All four call sites were repointed to the canonical taxonomy mapping; DDI evaluation was switched to consume expert ddi_binary labels rather than a derived quantity, and the malignant-probability computation was corrected from a single legacy class index to $\sum_{c\in\{0,1,2,3\}} p_c$.

Result. With the fix, DDI evaluation runs to completion for the first time, yielding – under the current checkpoint and the September 2026 artifact state (ddi_eval.csv regenerated 2026-09-05; figures from the superseded 2026-08-15 run, including AUC-ROC 0.6398 and an equalized-odds gap of 25.26%, are retired) – AUC-ROC 0.585 (95% CI 0.535–0.634, BCa), sensitivity 81.9%, specificity 24.1% at the 0.5 operating point on 656 expert-labelled held-out images, and an equalized-odds sensitivity gap of 25.96% (95% CI 12.67–41.80%) across tone groups: the first reliable fairness measurement obtained for this model, and one that demonstrably fails the <5% target.

V. Defect 2: Fabricated Skin-Tone Supervision

Deeper audit of the 25.96% gap traced its root cause upstream of evaluation, into training data. The ITA°skin-tone estimator was invoked without its required lesion-mask argument; every call raised an exception, silently caught and replaced with a hardcoded fallback (Fitzpatrick III). Combined with a dataset-loader clamp that mapped any missing tone value to the same fallback, 36,542 of 37,644 training rows (97%) carried the fabricated label. The Fitzpatrick auxiliary head and equalized-odds loss term were therefore optimized against a near-constant target, explaining why the model had no gradient signal to reduce cross-tone disparity despite an explicit fairness objective.

Fix. The estimator call was corrected to pass a lesion mask from Otsu segmentation, with unresolvable estimates marked as an explicit unknown label (-1) rather than defaulted; the dataset clamp was removed; loss and metric functions were updated to mask/exclude unknown-tone rows rather than treat them as a valid group. All 36,150 affected rows were re-estimated. The resulting tone distribution (Table I) is genuinely heterogeneous, if still light-skewed, reflecting the true composition of HAM10000/ISIC.

TABLE I
Training-set Fitzpatrick tone distribution before and after correction

Tone	Fabricated (before)	Estimated (after)
I	153	235
II	876	24,722
III	36,542	12,454
IV	62	220
V	10	12
VI	1	1

We deliberately excluded the 656 DDI samples from training supervision to preserve the integrity of the held-out fairness evaluation, and note DDI's coarser 3-bucket tone granularity as an additional reason against mixing it into 6-way tone supervision.

VI. Defect 3: INT8 Quantization Collapse via GeM Pooling

Post-training full-INT8 quantization, required to meet the < 6 MB on-device footprint, reduced top-1 accuracy from 73.7% (PyTorch, float32) to 3.3% on real test images – despite float32 ONNX and TFLite exports matching PyTorch exactly, ruling out a graph-conversion error. Inspecting per-tensor quantization scales isolated the cause to the GeM pooling operation, $y = (\sum_i x_i^p)^{1/p}$, with a learned exponent $p \approx 2.88$. The x^p intermediate spans activation magnitudes on the order of 10^6, forcing an INT8 quantization scale of 13,615 (versus ∼1–5 for ordinary convolutional activations) and collapsing essentially all post-pooling precision.

Fix. We adopted dynamic-range quantization (INT8 weights, float32 activations), which keeps the GeM power operation in full precision while still compressing weight storage. The resulting model is 5.51 MB, under the 6 MB budget.

Validation. On 150 real held-out test images against ground truth at the clinical decision threshold, the dynamic-range model is statistically indistinguishable from the PyTorch reference (Table II); on 100 images, ONNX matches PyTorch exactly and TFLite matches on 94% of top-1 predictions (99% of binary clinical decisions), with all disagreements confined to low-confidence, near-tied predictions.

We further found the validation script itself was a source of false alarms: parity checks on synthetic Gaussian-noise

TABLE II
PyTorch vs. deployed (dynamic-range TFLite) model at clinical threshold 0.25

Metric	PyTorch	Deployed (5.51 MB)
Decision accuracy	73.3%	72.0%
Sensitivity	90.9%	89.1%
Specificity	63.2%	62.1%
Top-1 accuracy	76.0%	74.7%

inputs reported only 22% agreement, but this reflects GeM pooling collapsing to near-uniform output probabilities under out-of-distribution noise in the reference model as well, not a quantization defect. We revised the validation protocol to use real held-out images by default.

VII. Clinical Threshold Calibration

With defects corrected, we recomputed the clinical operating threshold under a $\geq 96\%$ malignant-class sensitivity constraint, sweeping the summed malignant probability $\sum_{c\in\{0,1,2,3\}} p_c$ against the test set. The calibrated threshold is 0.25 (sensitivity 96.0%, specificity 55.4%, false-negative rate 3.97%, test-set AUC 0.9239, matching the shipped threshold_config.json), replacing an uncalibrated 0.5 placeholder and reducing missed malignant cases by 77% relative to the placeholder threshold.

VIII. Score-Level Analysis of the Out-of-Domain Failure

Threshold equalization (Section VIII-A) showed that no per-group threshold policy closes the DDI sensitivity gap. Score-level analysis explains why: the malignant probability itself barely separates benign from malignant lesions on DDI, and the deficit is worst exactly where the fairness gap is largest. All statistics derive from the audited prediction cache of the deployed checkpoint; CIs are (group × label) stratified percentile bootstrap (2,000 replications, seed 42) with raw n beside every estimate.

A. Threshold Equalization Does Not Close the Gap

The per-group experiment: DDI was split into (group × label) stratified tune (326) and eval (330) halves (85/86 malignant), per-group decision thresholds were fitted on the tune half only (the largest threshold meeting a $\tau \geq 0.95$ tune-half sensitivity target, τ chosen to minimize the tune-half EO gap), and the result was evaluated once on the held-out eval half against two controls: (A) the shipped uniform 0.25 threshold and (B) a uniform threshold re-tuned on the tune half (0.2947), which isolates any group-specific benefit from plain re-tuning. Eval-half outcomes: at the shipped threshold (A), sensitivity is 1.000/1.000/0.958 (I–II/III–IV/V–VI) with the EO gap at 0.042 [0.000, 0.125] – already below the 5% target at the point estimate, but only because specificity is negligible in every group (overall 0.016); there is no sensitivity gap left to close at this operating point. This is the same operating point at which Section IV reports the full-panel EO gap of 25.96%: the near-perfect sensitivity and the wide gap are two readings of one fact – malignant scores in every tone group sit almost always above threshold on DDI – and the narrowness of that margin is exactly what re-tuning cannot exploit without collapsing sensitivity. Re-tuning cannot buy specificity instead: uniform re-tuning (B) yields overall specificity 0.025, and the per-group config (C) reaches only 0.045 overall – its V–VI specificity of 0.087 [0.025, 0.163] is bought with V–VI sensitivity falling to 0.875, and the EO gap widens to 0.125 [0.000, 0.250]. The next subsection shows why: no threshold geometry can equalize what the score distributions do not rank.

B. No Operating Point Exists: Per-Group ROC Ceilings

TABLE III
Per-group ROC ceilings on the DDI panel (malignant score).

Group	n (pos/neg)	AUC [95% CI]	Sens@Spec≥0.6	Sens@Spec≥0.8
FST I–II	208 (49/159)	0.574 [0.486, 0.667]	0.490	0.224
FST III–IV	241 (74/167)	0.668 [0.592, 0.743]	0.662	0.392
FST V–VI	207 (48/159)	0.484 [0.382, 0.585]	0.396	0.229
ALL	656 (171/485)	0.585 [0.534, 0.636]	0.503	0.357

For FST V–VI the 95% CI on AUC includes 0.5 (Table III): the malignant score carries no measurable benign/malignant ranking signal for dark-skin images. This formalizes the threshold-equalization negative result: the disparity lives in the score distributions, not in threshold placement, and no post-hoc decision rule can recover performance the scores do not contain.

C. A Spurious Actinic-Keratosis Response Drives the False Positives

On true-benign DDI images the actinic-keratosis (AK) column is the largest malignant contributor on 63.3% of images (mean probability 0.375, 90th percentile 0.803); all fifteen highest-scoring false positives have AK as their top contributor (per-image AK probability 0.91–0.97). The same column is nearly silent on benign in-domain images (mean 0.046 on the test panel) – an order-of-magnitude shift (one-sided KS = 0.76, $p = 1 \times 10^{-218}$) – while contributing normally on true in-domain malignancies (mean 0.163).

Grad-CAM analysis [1] with matched comparators bounds what attention maps can establish. The AK head's activation is diffuse in both settings (median top-1% CAM mass share 0.023 on DDI false positives vs 0.019 on true AK cases; Mann–Whitney $p = 0.11$), so diffuseness is a property of the head, not a signature of the spurious response. The AK map leans toward darker pixels in both settings (median CAM–luminance correlation -0.29 on false positives, -0.21 on true AK, vs $+0.11$ for the MEL column on true melanomas). We accordingly do not claim attention morphology distinguishes the spurious response; the evidence for spuriousness is the score-level distribution shift above together with its tone-invariance below.

A structured visual review of all forty-five overlays (three cohorts of fifteen, reviewed side-by-side in triads)

then tested the capture-cue hypothesis directly. It supports the field-wide reading: on the false positives the activation spreads across the skin field and does not focal-lock onto salient non-lesion objects. Manual inspection found paper rulers in frame in 8/15 false-positive images and surgical ink markings in 6/15, yet neither attracts elevated CAM mass (edge-band mass share 0.410 vs 0.377 on true AK, Mann–Whitney $p = 0.28$; hot-pixel luminance ratio 0.95 vs 0.96, $p = 0.93$).

Conversely, the in- domain melanoma comparators are predominantly circular dermoscopic captures with black surrounds (9 of 15 with corner/center luminance below 0.75, vs 1 of 15 false positives) while the MEL head remains lesion-localized (edge-band share 0.244). Capture artifacts are therefore present in the false-positive cohort as expected, but the activation is not localized on them: the artifact census is context for the domain-shift finding, not its cause, and the spuriousness claim rests on the score-level evidence.

D. Domain Shift, Not Skin Tone, Drives the Artifact

Within DDI, benign-image AK probability is nearly flat across tone groups and highest for the lightest-tone group (FST I–II 0.440, FST V–VI 0.371, FST III–IV 0.318) – the opposite of a monotone tone gradient. At identical self-reported tone (FST I–II), DDI benigns show AK mean 0.440 vs test-panel 0.043 (KS = 0.78). We therefore frame the DDI disparity as a domain-generalization failure that disproportionately lands on the group with the weakest underlying ranking signal: the capture artifact and the tone-related performance gap compound rather than share a single cause.

E. A Post-Hoc Correction Cannot Repair the Score

A post-hoc down-weighting $p' = p_{\text{mal}} - \lambda P_{\text{AK}}$, with λ and the decision threshold fitted on one stratified half of DDI and evaluated once on the held-out half, yields at most marginal gains: tune-half AUC peaks at $\lambda^* = 0.25$ (0.609 vs 0.596 baseline) and degrades monotonically beyond. At λ^* the eval-half specificity is statistically unchanged ($0.025 \rightarrow 0.025$ overall) and the EO sensitivity gap moves from 0.125 [0.000, 0.250] to 0.083 [0.000, 0.208]. Aggressive deflation ($\lambda = 0.8$) destroys in-domain AK detection (AK-class recall at ≈80% specificity: $0.955 \rightarrow 0.444$) while still not fixing DDI specificity. The failure is structural: on eval-half benign images, deflating AK merely migrates the dominant malignant contributor to other columns (argmax share AK 62.3% → 28.3%, BCC 15.6% → 29.9%, MEL 16.4% → 23.0% at $\lambda = 0.8$) – single-column recalibration is whack-a-mole when the shared representation maps the capture domain itself onto malignancy. The deficit is therefore representational: closing it requires in-domain training signal (fine-tuning with DDI-style images, including hard negatives, and tone/capture augmentation), not score- or threshold-level correction.

IX. Discussion

A. Why Fairness Failures Were Silent

None of the three defects produced an exception during training or degraded top-line test accuracy (73.7% balanced accuracy was unaffected by any of them). Each was detectable only by (a) tracing a derived quantity back to its taxonomy source of truth, (b) inspecting the empirical distribution of an auxiliary label rather than trusting its presence, and (c) inspecting per-tensor quantization statistics rather than only end-to-end accuracy. We argue this generalizes: fairness and calibration code paths are exercised far less often than the primary training loop and are correspondingly more likely to harbor silent, high-impact bugs.

B. Retraining Alone Does Not Fix a Fairness Gap Rooted in Label Corruption

A naïve response to a 25% cross-tone fairness gap is to train longer. We show this would have been ineffective: the gap originated in corrupted supervision, not insufficient optimization – the label-choice bias mechanism documented by Obermeyer et al. [3], and a dataset-shift hazard of the kind catalogued by Finlayson et al. [4]. Correcting the label source is a prerequisite for any fairness-training intervention to have a measurable effect, and should precede, not follow, additional compute expenditure.

C. Domain Shift and the Limits of Post-Hoc Fairness Repair

The DDI analysis (Section VIII) adds a fourth lesson: post-hoc repairs (per-group thresholds, single-column recalibration) cannot fix a representational domain gap, and attention-map morphology alone cannot diagnose one – matched-comparator Grad-CAM analysis showed the diffuse activation previously read as an artifact signature is in fact the head's normal behavior. Given this representational deficit, we deliberately did not attempt in-domain fine-tuning in this work: Sections VII and VIII establish that every score- and threshold-level correction fails, so we identify targeted data acquisition (DDI-style capture-domain images, including benign hard negatives) and score-level fine-tuning under the corrected tone supervision as the appropriate future work, to be judged against the per-group ROC ceilings reported here.

D. Quantization Choice Is Architecture-Dependent

GeM pooling's learned exponent is a known source of quantization sensitivity; our finding that it produces an activation scale roughly 10^3-$10^4\times$ larger than typical convolutional layers offers a concrete diagnostic (per-tensor scale inspection) for detecting this class of failure before it reaches deployment, and dynamic-range quantization as a practical mitigation when full INT8 is required elsewhere in a pipeline.

E. Limitations

The post-correction tone distribution remains heavily skewed toward Types II–III (98.8% of training data), reflecting the true composition of HAM10000/ISIC 2019 rather than a residual bug. We expect the equalized-odds gap to narrow with genuine supervision but not to close without additional dark-skin-tone training data; closing it fully is a data-acquisition problem, not a modeling one. Best validation balanced accuracy (67.8–73.7% across evaluation protocols) had not converged at the time of writing; retraining to convergence under corrected tone supervision, and re-measuring the DDI fairness gap, is ongoing work. Two caveats attach to the Step VIII analysis. First, the AK head's preference for darker-than-scene pixels (observed identically on true AK cases) is left as an open question: it may reflect genuine clinical signal, since actinic damage correlates with sun-exposure appearance, or a confound with lesion-border shadowing; disentangling the two requires pixel-level attribution beyond the scope of this audit. Second, the visual overlay review was performed by the authors rather than an independent dermatologist, so the ruler and artifact counts are manual tallies pending expert confirmation.

X. Conclusion

We identified and corrected three independent, silently-compounding defects in a fairness-aware skin lesion classifier: a stale binary-label mapping, fabricated skin-tone training supervision from a swallowed exception, and a catastrophic INT8 quantization failure caused by a learned GeM pooling exponent. Each defect was invisible to standard top-line accuracy monitoring. We provide a corrected, on-device-validated deployment (5.51 MB, clinically calibrated threshold 0.25) and argue that fairness-critical ML pipelines require explicit data-lineage and quantization-scale auditing, not just held-out accuracy, before deployment claims can be trusted.

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